English

(1001CJA102121080)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PART TEST
28-01-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE-S, I(A), I & I(N)] : (SCORE-I)

PAPER-1

Time : 3 Hours **Maximum Marks : 198**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140,

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^9$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^8\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

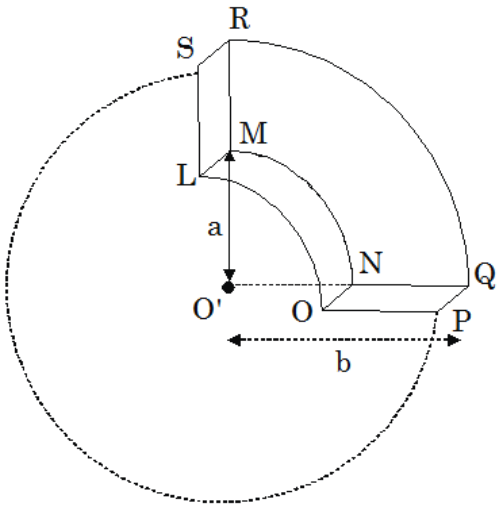
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. A conducting medium is shaped in a form of a quarter of an annulus of radii b and a (b > a) and thickness t. Specific resistance of the medium is ρ. Find the ratio of resistance between faces LMNO and PQRS to resistance between faces LMRS and OPQN.



(A) $\frac{2}{\pi^2}\left[\ln\left(\frac{b}{a}\right)\right]^2$

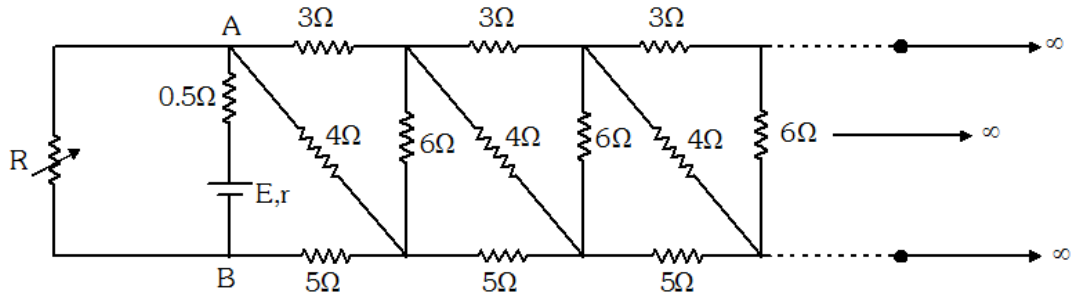
(B) $\frac{4(b-a)}{\pi^2 t}\left[\ln\left(\frac{b}{a}\right)\right]^2$

(C) $\frac{4}{\pi^2}\left[\ln\left(\frac{b}{a}\right)\right]^2$

(D) $\frac{4t}{\pi^2(b-a)}\left[\ln\left(\frac{b}{a}\right)\right]^2$

Space for Rough Work

2. Maximum rate of chemical energy consumed by the battery in watt is P_0. Find $\frac{P_0}{10}$. (EMF of the battery is E = 10V and its internal resistance is r = 0.5 Ω). R is choosen such that rate of chemical energy consumed by battery is maximum.



(A) 10 (B) 15 (C) 12 (D) 20

3. The main scale of vernier callipers reads in millimetre and its vernier is divided into 8 divisions, which coincide with 5 divisions of main scale. When two jaws of instrument touch each other the zero of the vernier coincide with the zero of main scale. A rod is tightly placed along its length between both jaws, it is observed that the zero vernier scale lies just left to 36th division of main scale and fourth division of vernier scale coincide with the main scale. Then the measured value is –

(A) 3.66 cm (B) 3.55 cm (C) 3.65 cm (D) 3.56 cm

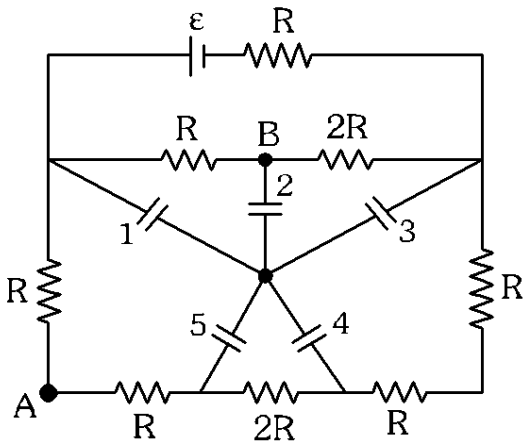
4. Complete the following calculations, reporting your answer to the correct number of significant figures :

$$\frac{10,569 - 958.82 - 1.8 \times 10^{-1}}{0.9950 + 8.60 + 0.015}$$

(A) 1000 (B) 1×10^3 (C) 1.00×10^3 (D) All of these

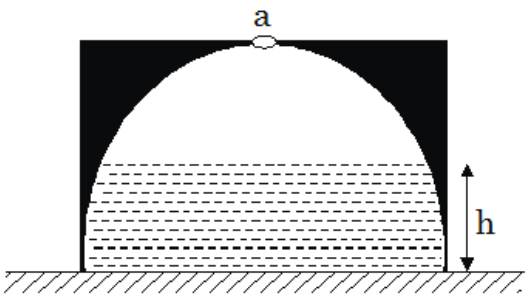
Space for Rough Work

5. Five identical capacitor (initially uncharged) are connected in circuit as shown in figure. Capacitance of each capacitor is C. In steady state charge on capacitor '1' is



(A) $\frac{14C\varepsilon}{45}$ (B) $\frac{4C\varepsilon}{9}$ (C) $\frac{2C\varepsilon}{3}$ (D) Zero

6. Consider a cuboidal vessel (2R × 2R × R) with a hemispherical cavity of radius R, kept at a horizontal smooth surface as shown in the figure. The vessel has very small hole of cross-sectional area "a". Now water of density ρ is poured into space developed due horizontal surface and vessel through hole very slowly. When the height of water level is h, the vessel lifts off the surface and liquid leaks through space generated. Find the value of h. (R = 2m, ρ = 10^3 kg/m^3, mass of vessel m = 9π kg); (Atmospheric pressure = 10^5 N/m^2).



(A) 20 cm (B) 15 cm (C) 25 cm (D) 30 cm

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

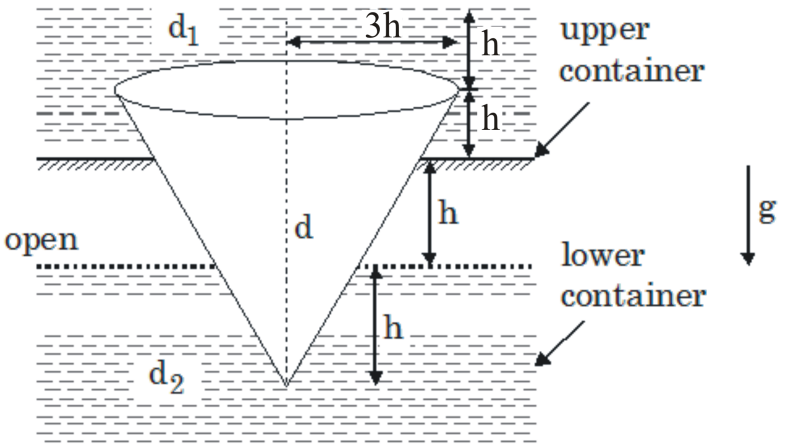
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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7. A solid cone of density d has $\frac{1}{3}$rd of its length in a fluid of density d_1, other $\frac{1}{3}$rd in air and the remaining $\frac{1}{3}$rd in a fluid of density d_2, as shown in figure. Given that $d_1 = \frac{3}{5}d$; the cone is in equilibrium in position shown with the contact force from the upper container's bottom being negligibly small, atmospheric pressure due to air is P_0 and force by atmospheric pressure on the middle part in air is F. Pick correct option(s)



(A) $d_2 = 10\ d$ (B) $F = P_0 \times 2\pi h^2$ (C) $F = P_0 \times 3\pi h^2$ (D) $d_2 = 30d$

Space for Rough Work

8. A water drop and a mercury drop is taken inside identical horizontal conical glass pipes. Then

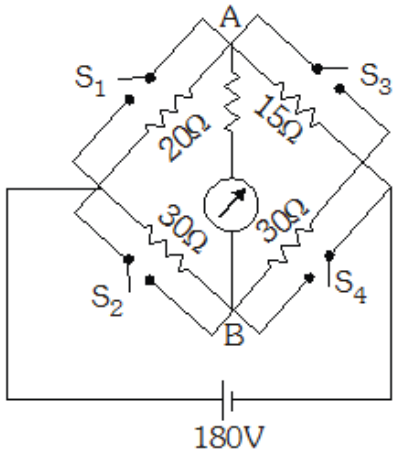
(A) The water drop tends to move towards the narrow end.

(B) The water drop tends to move towards the wide end.

(C) The mercury drop tends to move towards the wide end.

(D) The mercury drop tends to move towards the narrow end.

9. Consider the circuit shown. The resistance connected between the junction A and B is 60 Ω including the resistance of the galvanometer. The switches have no resistance when shorted and infinite resistance when opened. All the switches are initially open and they are closed as given in the options. Choose correct option(s) :



(A) If only switch S_1 is closed current through the battery is 20 Amp

(B) If only switch S_2 is closed then current in galvanometer from B to A

(C) If only switch S_3 is closed then current in galvanometer from B to A

(D) If only switch S_4 is closed then current in galvanometer from A to B

Space for Rough Work

10. We have five identical galvanometer of resistance R and five identical resistance R.
Situation → (1) all galvanometer and resistance are connected in series.
Situation → (2) all galvanometer connected in parallel, all resistance connected in parallel and combination of galvanometers and combination of resistances connected in series.
Situation → (3) all galvanometer connected in parallel, all resistance connected in series and combination of galvanometers and resistances are connected in series.
Situation → (4) all galvanometer connected in series, all resistance connected in parallel and combination of galvanometers and resistance connected in series.
The complete combination in all situation acts as voltmeter.

(A) Range of voltmeter in situation (1) is 5 times range in situation (2) and > range in situation (4)

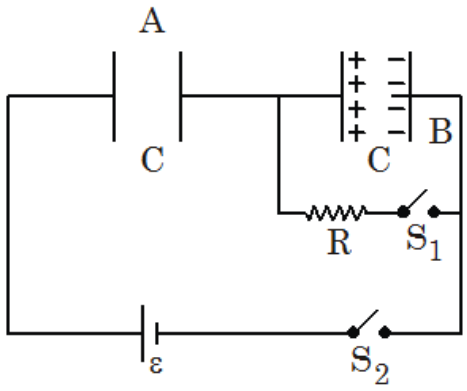
(B) Range of voltmeter in situation (1) is 5 times range in situation (2) and < range in situation (4)

(C) The value of range of voltmeter is maximum in the situation (3). (In above all given situation)

(D) Range of voltmeter in situation (3) is > range in situation (4)

Space for Rough Work

11. Two identical capacitor A and B of capacitance 'C' each connected in the circuit as shown in the figure with ideal battery. Capacitor B having charge $Q_0 = C\varepsilon$ initially. At t = 0 both S_1 and S_2 closed simultaneously.



(A) Current in R as function of time is $\frac{\varepsilon}{R}e^{-t/2RC}$

(B) Charge on capacitor A as function of time is $C\varepsilon(1 - e^{-t/2RC})$

(C) Charge on capacitor B in steady state is zero

(D) Total work done by the battery till steady state is reached is $2C\varepsilon^2$

12. The universal gravitational constant is measured experimentally using Cavendish's experiment. This experiment is a very precise experiment which gives the value of G as $(6.6742 \pm 0.0010) \times 10^{-11}$ N.m^2/kg^2. Since the universal gravitational constant has some error in it, using it if we find escape speed and acceleration due to gravity on Earth's surface, our values will have error in it. (Assume mass of Earth = 6.0000×10^{24} kg & radius of Earth = 6.4000×10^6 m.) What will be the maximum percentage error in calculating the escape speed on Earth ? (The measurement of mass of Earth has an error of 0.01 % and radius of Earth has an error of 0.02%):-

(A) 0.045% (B) 0.0225 % (C) 0.015 % (D) 0.0025%

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

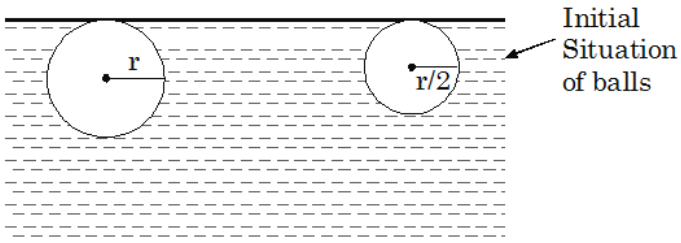
Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

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1. The pitch of screw gauge is 1 mm and its circular scale is divided into 50 divisions. When nothing is put between the studs 44th division of circular scale coincides with the reference line and circular scale lies between 3rd and 4th division of main scale. When a glass plate is placed between the studs, the circular scale lies between 13th and 14th division of main scale and circular scale reads 26 divisions. Find the thickness of the plate (in mm).

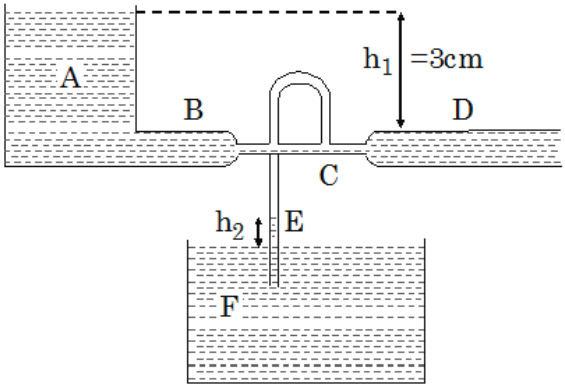
2. Two balls of radii r and $\frac{r}{2}$ are released inside a deep water tank. Their initial accelerations are found to be $\frac{g}{2}$ and $\frac{g}{4}$ downward respectively. If the velocity of smaller ball relative to the larger ball, a long time after the two balls are released is $\frac{x\, r^2 \rho g}{54\, \eta}$. Coefficient of viscosity is given to be η. Find x.

(ρ = density of water)

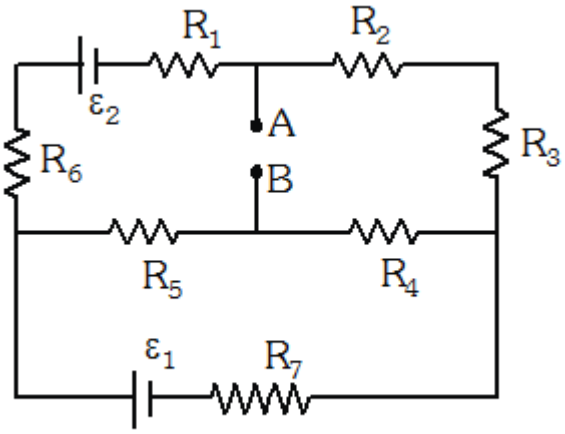


Space for Rough Work

3. Two vary large open tanks A and F contain the same liquid as shown in the figure. A horizontal pipe BCD having a constriction at C and open to the air at D, leads out of the bottom of tank A, and a vertical pipe E opens into the constriction at C and dips into the liquid in tank F. Assume streamline flow and no viscosity. If the cross sectional area at C is one half the area at D and if D is a distance 3 cm below the level of the liquid in A, upto what height h_2 (cm) will the liquid rise in pipe E? (Neglect surface tension)

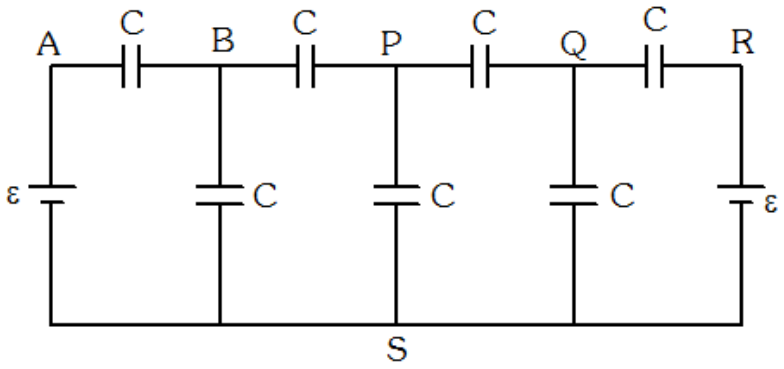


4. In the given circuit diagram, if ideal ammeter connected between point A and B its reading is 5A, if ammeter of resistance 3Ω is connected between A and B its reading is 3A. What will be reading of ideal voltmeter if it is connected between A and B (in volt)?



Space for Rough Work

5. In the given circuit diagram initially all capacitor are uncharged. Answer the following question in steady state. Heat developed during charging is $nC\varepsilon^2$. Find $\frac{1}{n}$.



Space for Rough Work

6. In the circuit shown (figure 1), all the switches are open initially and all capacitors are uncharged. Now first only S_1 is closed and then opened when steady state is reached. Then S_2 is closed and again opened when steady state is reached, after this S_3 is closed and opened when again steady state is reached, this process repeated countineously. After closing sixth switch, the fourth capacitor (8C) is removed and connected (say at t = 0) to an uncharged capacitor of original capacitance C which now has a dielectric (dielectric constant 2) filling half its volume (see figure 2), through a resistor R. If final charge on capacitor B when steady state reached in figure 2 is $\frac{8}{n}CV$. The value of n is

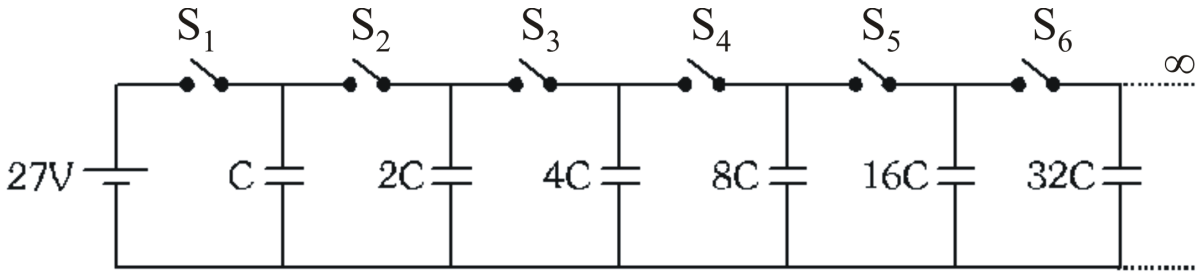


Fig.(1)

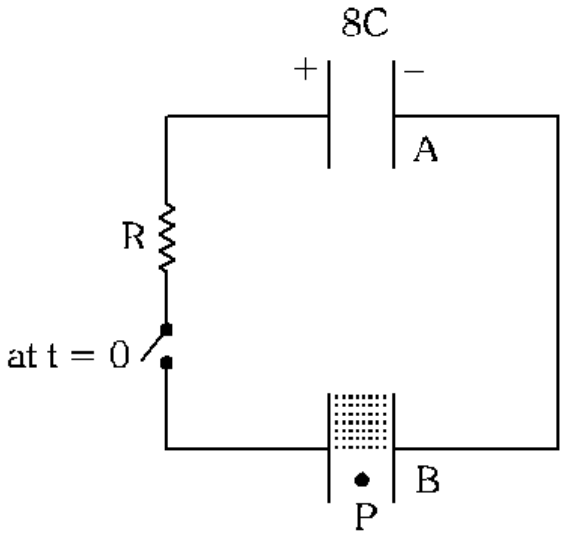


Fig. (2)

Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. Select incorrect statement wrt H (protium), D and T

(A) Both D & T can be used for radioactive–labelling in organic compound.

(B) Tritium decays by β–emission.

(C) All three can't be used for radioactive–labelling in organic compound.

(D) Half-life of T is approximate 12.3 years.

2. Which of the following pair of reagents can be used for producing hydrogen gas?
(I) Zn + dil. H_2SO_4 (II) Zn + NaOH (III) Al + HCl (g) (IV) Al + NaOH

(A) I, II and III only
(B) II, III and IV only
(C) I, III and IV only
(D) I, II, III and IV

Space for Rough Work

3. For a gaseous chemical reaction the value of $K_P = \frac{4}{3}atm^{-1}$ at 227ºC in a closed container. The correct statement for the reaction at same temperature is/are : (Given $R = \frac{1}{12}\frac{litre - atm}{k - mol}$)

(A) $K_C = \frac{20}{3}$ mole litre^{-1}

(B) If volume of container is increased equilibrium shifts in backward direction.

(C) If partial pressure of each reactant is doubled K_P becomes 4 times.

(D) Can't measure K_C for the reaction as data are insufficient.

4. What should the solubility of AgCN in acidic buffer solution of pH = 5.
$Ka_{(HCN)} = 5 \times 10^{-10}$; $K_{sp\,(AgCN)} = 6 \times 10^{-17}$

(A) 7.7×10^{-9} M

(B) 1.1×10^{-6} M

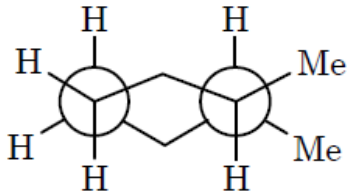
(C) 2.1×10^{-7} M

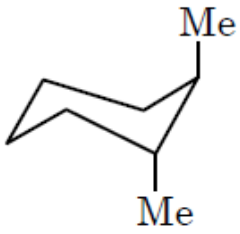
(D) 3.2×10^{-8} M

Space for Rough Work

5. The correct stability order of the following species is.

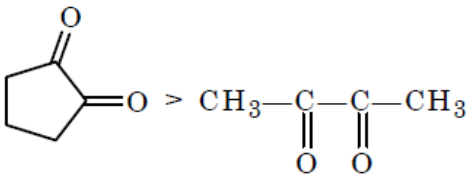
(a) 

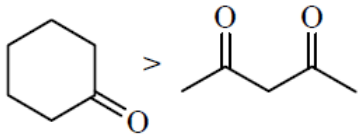
(b) 

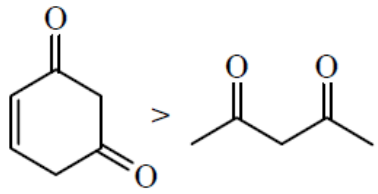
(c) 

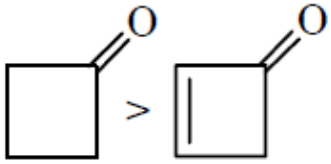
(A) c < a < b (B) c = b < a (C) c < a = b (D) a = b = c

6. Which of the following is incorrect for Enol % ?

(A) 

(B) 

(C) 

(D) 

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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7. Which of the following incorrect for stereoisomers 2-chloro-1-fluoro propene.

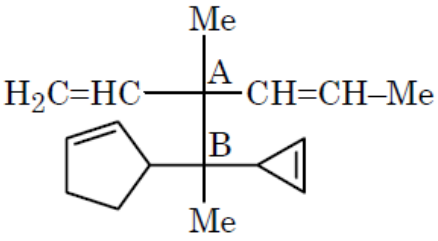
(A) E–isomer is having more water solubility then Z–isomer.

(B) Z–isomer is having more water solubility then E–isomer

(C) E & Z both isomers are equally water soluble

(D) E–isomer is having intramolecular H–bonding.

8. Correct R, S configuration for following compound is:



(A) R at A, R at B (B) R at A, S at B (C) S at A, R at B (D) S at A, S at B

Space for Rough Work

9. Identify the options which is/are correct w.r.t properties of elements.

(A) IE of Li^- will be more than Na^-

(B) Inspite of lower charge, the ionic potential (charge/size ratio) of Li^+ is more than that of Mg^{2+}

(C) Cs is the best reducing agent, therefore among alkali metal, reducing power is influenced by hydration energy.

(D) Correct order of EN of 13th group elements is B > Tℓ > In > Ga > Al.

10. Match the following.

In List–I the comparision of IE values and EA values of X and Y are given and in List–II.The elements X and Y are listed. Match them.

	List I		List II	
			X	Y
(P)	IE_1 of X is more than that of Y	(1)	Be	B
(Q)	IE_2 of X is more than that of Y	(2)	B	C
(R)	IE_3 of X is more than that of Y	(3)	C	N
(S)	EA of X is less than that of Y	(4)	N	O

(A) P → 1,4; Q → 2; R → 1,3; S → 1,2,4

(B) P → 1,2,4; Q → 2; R → 1,3; S → 1,2,4

(C) P → 1,3; Q → 1,2; R → 1,3; S → 1,2,4

(D) P → 2,3; Q → 1,2; R → 1,3; S → 1,2,4

Space for Rough Work

11. Following two equilibria are simultaneously established in a container
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2$; $CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$
If some Ni(s) is introduced in the container establishing another equilibrium.
Which among the following statements is/are definitely true about this system ?

(A) $COCl_2$ concentration decreases as the new equilibrium is established.

(B) As Ni is a solid it will not have any impact on the existing two equilibrium reactions.

(C) PCl_3 concentration decreases to re–attain the equilibrium.

(D) PCl_5 concentration increases to attain the new equilibrium.

12. Solubility product of AgCl (s) in water at 25°C is 12×10^{-10} M^2 then select the correct options.

(A) Solubility of AgCl in presence of 10^{-5} M $AgNO_3$ is 3×10^{-5} moles/L.

(B) Solubility of AgCl in presence of excess of KCl will increase due to formation of complex $AgCl_2^-$ (aq.)

(C) Solubility of AgCl in presence of 2M NH_3 is greater than that in pure water.

(D) Solubility of AgCl remains unaffected on adding any amount of NaBr because there is no common ion effect.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

---

1. 0.005 mol of H_2A^- and 0.105 mol A^{3-} are mixed in 1 litre solution. Find out the pH of final solution.
(H_3A $Ka_1 = 10^{-5}$, $Ka_2 = 10^{-9}$ $Ka_3 = 10^{-12}$)

2. $HCN \rightleftharpoons H^+ + CN^-$; $K = 10^{-9}$

$NH_4OH \rightleftharpoons NH_4 + OH^-$; $K = 10^{-5} \times 2$

$HCN + KOH \rightarrow KCN + H_2O$; K_1

$NH_4OH + HCN \rightarrow NH_4CN + H_2O$; K_2

$NH_4NO_3 + H_2O \rightarrow NH_4OH + HNO_3 + K_3$

$\frac{K_1 \times K_3}{K_2} = x \times 10^{-7}$; 'x' is :

Space for Rough Work

3. In how many cases electron gain enthalpy is negative.

(i) $H(g) + e \rightarrow H^-(g)$

(ii) $Li(g) + e \rightarrow Li^-(g)$

(iii) $Na(g) + e \rightarrow Na^-(g)$

(iv) $K(g) + e \rightarrow K^-(g)$

(v) $P(g) + e \rightarrow P^-(g)$

(vi) $O(g) + e \rightarrow O^-(g)$

(vii) $O^-(g) + e \rightarrow O^{2-}(g)$

(viii) $N(g) + e \rightarrow N^-(g)$

(ix) $Ar(g) + e \rightarrow Ar^-(g)$

4. How many incorrect statements ?

(a) CO which is a major pollutant resulting from the combustion of fuels in automobiles plays a major role in photochemical smog.

(b) Classical smog has an oxidising character while the photochemical smog is reducing in character.

(c) Photochemical smog occurs in day time whereas the classical smog occurs in the morning hours.

(d) During formation of smog the level of ozone in the atmosphere goes down.

(e) Classical smog is good for health but not photochemical smog.

(f) Chlorofluorocarbons are responsible for ozone layer depletion.

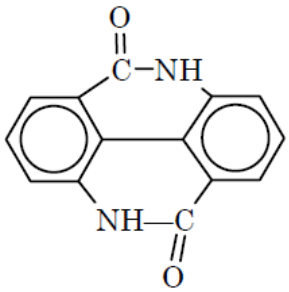
(g) Green house effect is responsible for global warming.

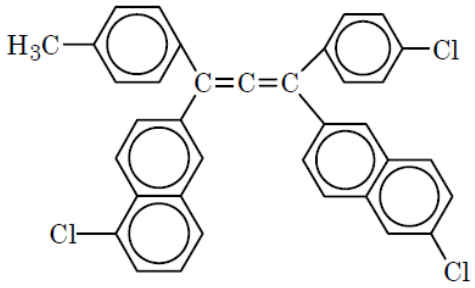
(h) Ozone layer does not permit infrared radiation from the sum to reach the earth.

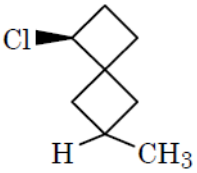
(i) Acid rain is mostly because of oxides of nitrogen and sulphur.

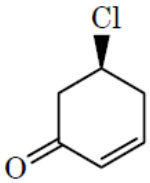
Space for Rough Work

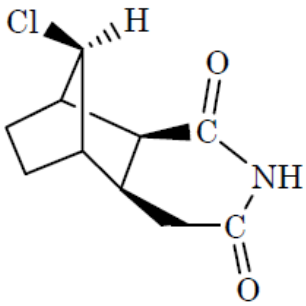
5. How many of the following compounds are chiral ?

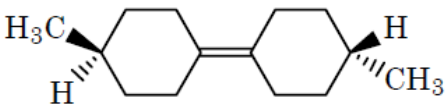
(i) 

(ii) 

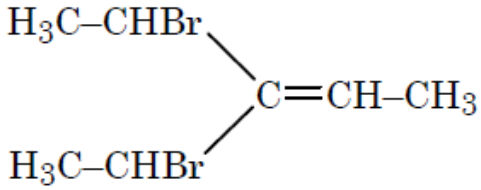
(iii) 

(iv) 

(v) 

(vi) 

Space for Rough Work

6. 

For above compound's

P → total number of stereoisomers.

Q → total number of enantiomeric pairs

R → total number of meso compounds

Then give your answer in 3 digit's as PQR.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

---

1. The interval to which b must belong so that the function $f(x) = \left(1 - \frac{\sqrt{21 - 4b - b^2}}{b+1}\right) x^3 + 5x + \sqrt{6}$ is increasing at every point of its domain is

(A) $[-10, -1)$ (B) $[-8, -6]$ (C) $[2, 3]$ (D) $[3, 8]$

2. If $\int u\frac{d^2v}{dx^2}dx = u\frac{dv}{dx} - v\frac{du}{dx} + w$ then w is equal to :

(A) $\int v\frac{d^2u}{dx^2}dx$ (B) $\int u\frac{d^2v}{dx^2}dx$

(C) $\int v\left(\frac{du}{dx}\right)^2 dx$ (D) $\int u\left(\frac{dv}{dx}\right)^2 dx$

3. Which of the following equation has real solution(s) ?

(A) $\cos(\sin x) = \frac{1}{2}$ (B) $(x^2 + 1)^{10} = 2x - x^2 - 2$

(C) $(3 + \cos x)^2 = 4 - 2\sin^2 x$ (D) $\sin x + \cos x = \frac{\pi}{2}$

Space for Rough Work

4. There is a point P inside an equilateral triangle ABC of side d whose distances from the vertices A,B & C are 3, 4, 5 respectively. Rotate the triangle through 60° about C. Let A goes to A' and P to P'. The area of triangle PAP' is

(A) 8 (B) 12 (C) 6 (D) None of these

5. Tangent is drawn to ellipse $\frac{x^2}{27} + y^2 = 1$ at $(3\sqrt{3}\cos\theta, \sin\theta)$ $\left(\text{where } \theta \in \left(0, \frac{\pi}{2}\right)\right)$. Then the value of θ such that sum of intercepts on axes made by this tangent is least, is

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{4}$

6. If $f : R \to R$ is a function such that $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3)$ for $x \in R$ then the value of $f(2)$ is

(A) 5 (B) 10 (C) 6 (D) -2

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

7. If a, b and c be in A.P. Then the correct results from the given statements are :

(A) $\cot\frac{A}{2}, \cot\frac{B}{2}$ and $\cot\frac{C}{2}$ are in A.P.

(B) $\cos A \cot\frac{A}{2}, \cos B \cot\frac{B}{2}$ and $\cos C \cot\frac{C}{2}$ are in A.P.

(C) $a\cos^2\frac{C}{2} + c\cos^2\frac{A}{2} = \frac{3b}{2}$

(D) $\cot\frac{A}{2}\cot\frac{C}{2} = 2$

Space for Rough Work

8. If $y = mx + 5$ is a tangent to the curve $x^3y^3 = ax^3 + by^3$ at P(1, 2), then

(A) $a + b = \frac{19}{5}$ (B) $a > b$ (C) $a < b$ (D) $a + b = \frac{5}{18}$

9. $\frac{\alpha^2}{1 - \tan^2 x} = \frac{\sin^2 x + \alpha^2 - 2}{\cos 2x}$ has a solution if

(A) $\alpha \in (-\infty, -1) \sim \{-\sqrt{3}\}$ (B) $\alpha \in (1, \infty) \sim \{\sqrt{3}\}$

(C) $\alpha \in (-1, 1)$ (D) α is any real number

10. Let $f(x) = \int_0^x e^{-t^2}(t-5)\left(t^2 - 7t + 12\right) dt$ for all $x \in (0, \infty)$, then

(A) f has a local maximum at x = 4 and a local minimum at x = 3.

(B) f is decreasing on $(3, 4) \cup (5, \infty)$ and increasing on $(0, 3) \cup (4, 5)$.

(C) There exists atleast two $c \in (0, \infty)$ such that $f''(c) = 0$

(D) There exists some $c \in (0, \infty)$ such that $f'''(c) = 0$

Space for Rough Work

11. Let $f(x) = e^{ax} \sin(bx + c)$ and $f''(x) = r^2 e^{ax} \sin(bx + \theta)$, then (a,b,c,r and θ are real positive arbitrary constants and $b < a$)

(A) $r = a^2 + b^2$

(B) $r = \sqrt{a^2 + b^2}$

(C) $\theta = c + 2\tan^{-1}\left(\frac{b}{a}\right)$

(D) $\theta = 2\text{atan}^{-1}\left(\frac{b}{a}\right)$

12. Let f be a continuous function and $f''(x) < 0 \ \forall \ x \in R$. Let $f(0) = 1$ and $f(1) = 0$. Which of the following statements are necessarily true ?

(A) There exists a number k in [0, 1] such that $f(k) \geq f(x)$ for all x in [0, 1]

(B) $f'(1) < -1$

(C) There exists a number k in [0, 1] such that $f(k) = \frac{1}{2}$

(D) $f'(0) > -1$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

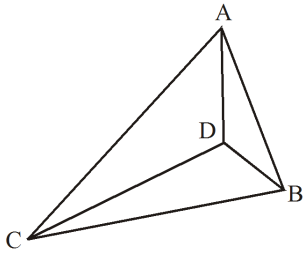
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- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks : 0 In all other cases.

---

1. The number of solutions that the equation sin (cos (sin x)) = cos (sin (cos x)) has in $\left[0, \frac{\pi}{2}\right]$ is :

2. In the adjoining figure, AC = BC, ∠ACB = 96° and D is a point inside ΔABC such that ∠DAB = 18°, ∠DBA = 30°. Then the value of $\left[\frac{\angle ADC}{17}\right]$ is



(where [.] denotes greater integer function)

3. Let y = f(x) be a one-one function satisfying the equation $\frac{d^2y}{dx^2} + \frac{d^2x}{dy^2} = 0$ and f'(0) = 1, f(1) = 2, then f(0) is :

Space for Rough Work

4. If product of all possible values of k for which angle of intersection of curves $y = e^{3x}$ and $y = k^x$ is 45° is λ then $|\ell n\lambda|$ is equal to

5. If the equation of tangent drawn to curve $y = f(x)$ at point P(3, 5) is $5x - 4y + 5 = 0$ and $\lim\limits_{x\to 3} \dfrac{\left(3^{4f(x)} - 2\left(1+3+3^2+\ldots\ldots+3^{19}\right)-1\right)^2}{1-\cos(\ln(4-x))} = 2(a\ 3^{20}\ \ell n\ c)^2$ (where a and c are prime numbers), then a + c is

6. Let $f : R \to [-8, 8]$ be an onto function which is given by $f(x) = \dfrac{bx}{(a-3)x^3 + x^2 + 4}$ $a, b, \in R^+$. If the set of values of m for which the equation $f(x) = mx$ has 3 distinct real solution is (p, q), then the value of (a + b + p + q) is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work

(1001CJA102121081)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)
PART TEST
28-01-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE-S, I(A), I & I(N)] : (SCORE-I)

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY/कृपया इन निर्देशों को ध्यान से पढ़ें

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics**, **Chemistry** and **Mathematics** having **30 questions** in each subject and each subject having **Two sections.**
   (i) Section-I contains 20 **multiple choice** questions with **only one correct** option.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
   (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
   Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writting particulars/marking responses on **Side–1** and **Side–2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/ Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. **Take g = 10 m/s² unless otherwise stated.**

महत्वपूर्ण निर्देश :

1. परीक्षा पुस्तिका के इस पृष्ठ पर आवश्यक विवरण *नीले/काले बॉल पाइंट पेन* से तत्काल भरें। *पेन्सिल का प्रयोग बिल्कुल वर्जित हैं।*
2. परीक्षार्थी अपना फार्म नं. (निर्धारित जगह के अतिरिक्त) परीक्षा पुस्तिका/उत्तर पत्र पर कहीं और न लिखें।
3. इस परीक्षा पुस्तिका में **90** प्रश्न हैं।
4. इस परीक्षा पुस्तिका में तीन भाग 1, 2, 3 हैं, जिसके प्रत्येक भाग में **भौतिक विज्ञान**, **रसायन विज्ञान** एवं **गणित** के **30 प्रश्न** हैं और प्रत्येक विषय में 2 खण्ड है।
   (i) खण्ड-I में 20 **बहुविकल्पीय** प्रश्न है। जिनके **केवल एक** विकल्प सही हैं।
   अंक योजना : +4 सही उत्तर के लिए, 0 प्रयास नहीं करने पर तथा –1 अन्य सभी अवस्थाओं में।
   (ii) **खण्ड-II में 10 संख्यात्मक मान प्रकार के प्रश्न है।** किन्ही 5 प्रश्नों का उत्तर दीजिए। किये गये प्रश्नों में से केवल प्रथम पाँच प्रश्नों को ही अंक दिये जायेंगे।
   अंक योजना : +4 सही उत्तर के लिए तथा 0 अन्य सभी अवस्थाओं में।
5. उत्तर पत्र के **पृष्ठ–1** एवं **पृष्ठ–2** पर वांछित विवरण एवं उत्तर अंकित करने हेतु केवल **नीले/काले बॉल पाइंट पेन** का ही प्रयोग करें। **पेन्सिल का प्रयोग सर्वथा वर्जित है।**
6. परीक्षार्थी द्वारा परीक्षा कक्ष/हॉल में परिचय पत्र के अलावा किसी भी प्रकार की पाठ्य सामग्री मुद्रित या हस्तलिखित कागज की पर्चियों, मोबाइल फोन या किसी भी प्रकार के इलेक्ट्रानिक उपकरणों या किसी अन्य प्रकार की सामग्री को ले जाने या उपयोग करने की अनुमति नहीं हैं।
7. रफ कार्य परीक्षा पुस्तिका में केवल निर्धारित जगह पर ही कीजिये।
8. परीक्षा समाप्त होने पर, परीक्षार्थी कक्ष/हॉल छोड़ने से पूर्व उत्तर पत्र कक्ष निरीक्षक को अवश्य सौंप दें। **परीक्षार्थी अपने साथ इस परीक्षा पुस्तिका को ले जा सकते हैं।**
9. **उत्तर पत्र को न मोड़ें एवं न ही उस पर अन्य निशान लगाऐं।**
10. **g = 10 m/s² प्रयुक्त करें, जब तक कि अन्य कोई मान नहीं दिया गया हो।**

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR/निरीक्षक के अनुदेशों के बिना मुहरें न तोड़ें

Name of the Candidate (in Capitals) _____
परीक्षार्थी का नाम (बड़े अक्षरों में) :

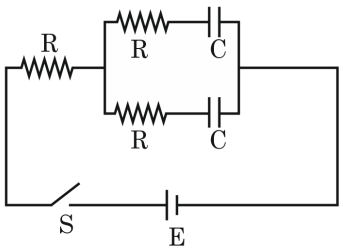
Form Number : in figures _____
फॉर्म नम्बर : अंकों में
: in words _____
: शब्दों में

Centre of Examination (in Capitals) : _____
परीक्षा केन्द्र (बड़े अक्षरों में) :

Candidate's Signature : _____ Invigilator's Signature : _____
परीक्षार्थी के हस्ताक्षर : निरीक्षक के हस्ताक्षर :

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14. The switch S in the circuit diagram figure is closed at t = 0. The charge on each capacitor at any time t is



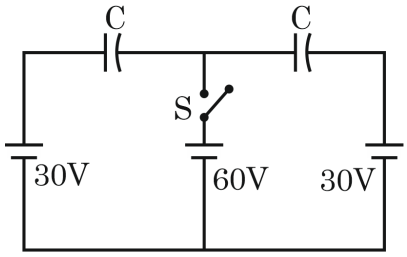
(A) $q_{(t)} = EC\,(1 - e^{-2t/3RC})$

(B) $q_{(t)} = EC\,(1 - e^{-t/2RC})$

(C) $q_{(t)} = EC\,(1 - e^{-t/3RC})$

(D) $q_{(t)} = EC\,(1 - e^{-3t/2RC})$

15. Find heat dissipated in the circuit after switch S is closed. C = 2 μF.



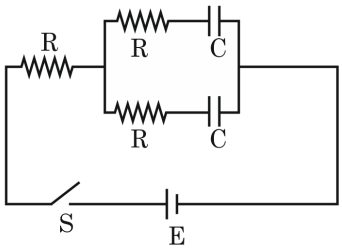
(A) 1.8 mJ

(B) 1.9 mJ

(C) 1.6 mJ

(D) 180 mJ

14. प्रदर्शित परिपथ में स्विच S को t = 0 पर बंद कर दिया जाता है। किसी समय t पर प्रत्येक संधारित्र पर आवेश है :-



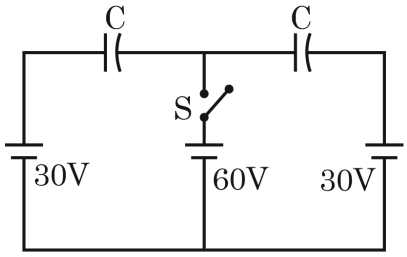
(A) $q_{(t)} = EC\,(1 - e^{-2t/3RC})$

(B) $q_{(t)} = EC\,(1 - e^{-t/2RC})$

(C) $q_{(t)} = EC\,(1 - e^{-t/3RC})$

(D) $q_{(t)} = EC\,(1 - e^{-3t/2RC})$

15. स्विच S को बंद करने के पश्चात् परिपथ में व्ययित ऊष्मा ज्ञात कीजिये। C = 2 μF.



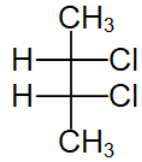
(A) 1.8 mJ

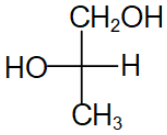
(B) 1.9 mJ

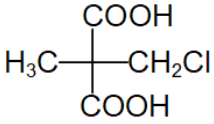
(C) 1.6 mJ

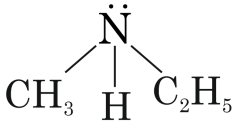
(D) 180 mJ

18. Which of the following compound is resolvable?

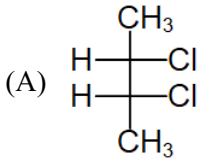
(A) 

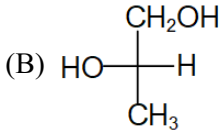
(B) 

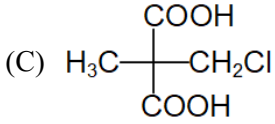
(C) 

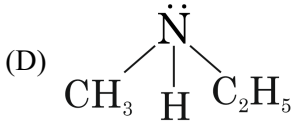
(D) 

18. निम्न में से कौन सा यौगिक पृथक्करणीय है?

(A) 

(B) 

(C) 

(D) 

19. $M(OH)_2(s)$ is involved in the following two equilibrium

$M(OH)_2(s) \rightleftharpoons M^{2+}(aq) + 2OH^-(aq)$

$K_{sp} = 1.2 \times 10^{-17}$

$M(OH)_2(s) + 2OH^- \rightleftharpoons [M(OH)_4]^{2-}(aq)$ $K_f = 0.12$

Then concentration of OH^- at which solubility of $M(OH)_2$ is minimum

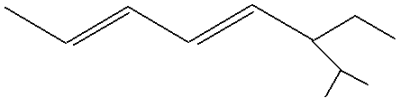
(A) 10^{-4}

(B) 1.2×10^{-9}

(C) 2.18×1010^{-6}

(D) None of these

20. The total stereoisomers for the following compound are ?



(A) 8

(B) 7

(C) 6

(D) 10

19. निम्न दो साम्यों में $M(OH)_2(s)$ सम्मिलित है

$M(OH)_2(s) \rightleftharpoons M^{2+}(aq) + 2OH^-(aq)$

$K_{sp} = 1.2 \times 10^{-17}$

$M(OH)_2(s) + 2OH^- \rightleftharpoons [M(OH)_4]^{2-}(aq)$ $K_f = 0.12$

तो OH^- की वह सान्द्रता जिस पर $M(OH)_2$ की विलेयता न्युनतम है

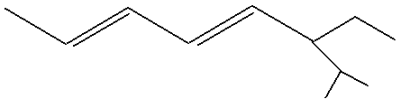
(A) 10^{-4}

(B) 1.2×10^{-9}

(C) 2.18×1010^{-6}

(D) इनमें से कोई नहीं

20. निम्न यौगिक के कुल त्रिविम समावयवी हैं ?



(A) 8

(B) 7

(C) 6

(D) 10

4. Let ABCDEFGHIJKL be a regular dodecagon. Then the value of $\frac{AB}{AF}+\frac{AF}{AB}$ is equal to _____.

(A) 4 (B) 2 (C) 6 (D) 3

5. If there is an error of α % in measuring the edge of a cube, then the percent error in estimating its volume is

(A) α (B) 3α

(C) $\frac{\alpha}{3}$ (D) α^3

6. In a triangle ABC, $2a^2 + 4b^2 + c^2 = 4ab + 2ac$, then numerical value of cos B is equal to

(A) 0 (B) $\frac{-1}{8}$

(C) $\frac{1}{8}$ (D) $\frac{7}{8}$

7. If $y = \frac{1}{x}$, then value of $\frac{dy}{\sqrt{1+y^4}}+\frac{dx}{\sqrt{1+x^4}}$ is equal to -

(A) 0 (B) 2

(C) 4 (D) –2

8. Number of solutions of the inequation $\sqrt{2-\cos^{19}x} \le \sin^{15}x + \cos^{23}x$ in the interval $[-2\pi, 2\pi]$

(A) 0 (B) 1 (C) 2 (D) 3

4. माना ABCDEFGHIJKL सम द्वादश भुज है। तब $\frac{AB}{AF}+\frac{AF}{AB}$ का मान होगा

(A) 4 (B) 2 (C) 6 (D) 3

5. यदि किसी घन की भुजा को मापने में α % की त्रुटि है, तो उसके आयतन का अनुमान लगाने में प्रतिशत त्रुटि होगी

(A) α (B) 3α

(C) $\frac{\alpha}{3}$ (D) α^3

6. त्रिभुज ABC में, $2a^2 + 4b^2 + c^2 = 4ab + 2ac$ हो, तो cos B का संख्यात्मक मान होगा

(A) 0 (B) $\frac{-1}{8}$

(C) $\frac{1}{8}$ (D) $\frac{7}{8}$

7. यदि $y = \frac{1}{x}$ हो, तो $\frac{dy}{\sqrt{1+y^4}}+\frac{dx}{\sqrt{1+x^4}}$ का मान होगा -

(A) 0 (B) 2

(C) 4 (D) –2

8. अन्तराल $[-2\pi, 2\pi]$ में असमिका $\sqrt{2-\cos^{19}x} \le \sin^{15}x + \cos^{23}x$ के हलों की संख्या होगी

(A) 0 (B) 1 (C) 2 (D) 3

9. The value of
$\lim_{x\to 0} \frac{e\ln(\sec(ex)\sec(e^2x).....\sec(e^{40}x)}{(e^x+2x)^{1/x}\sin^2 x}$
is $\frac{1}{2}\left[\frac{e^a-1}{e^b-1}\right]$, then the value of a + b
(where a, b ∈ N)

(A) 84 (B) 42

(C) 82 (D) 40

10. Given a right circular cone has a volume V_1 and the largest right circular cylinder that can be inscribed in the cone has a volume V_2. Then $V_1 : V_2$ is $\frac{a}{b}$ (where a, b are coprime). Find the value of $a^2 - b^2$.

(A) 25

(B) 65

(C) 64

(D) 20

11. If c is a point at which Rolle's theorem holds for the function, $f(x) = \log_e\left(\frac{x^2+\alpha}{7x}\right)$ in the interval [3, 4], where $\alpha \in R$, then f ″(c) is equal to :

(A) $\frac{\sqrt{3}}{7}$ (B) $\frac{1}{12}$

(C) $-\frac{1}{24}$ (D) $-\frac{1}{12}$

9. $\lim_{x\to 0} \frac{e\ln(\sec(ex)\sec(e^2x).....\sec(e^{40}x)}{(e^x+2x)^{1/x}\sin^2 x}$
का मान $\frac{1}{2}\left[\frac{e^a-1}{e^b-1}\right]$ हो, तो a + b का मान होगा
(जहाँ a, b ∈ N)

(A) 84 (B) 42

(C) 82 (D) 40

10. दिया गया है एक समकोणीय शंकु जिसका आयतन V_1 तथा सबसे बडा समकोणीय बेलन, जो शंकु के अन्तर्गत है जिसका आयतन V_2 है। तब $V_1 : V_2$ बराबर $\frac{a}{b}$ (जहाँ a, b सह-अभाज्य) हो, तो $a^2 - b^2$ का मान ज्ञात कीजिए

(A) 25

(B) 65

(C) 64

(D) 20

11. यदि एक बिन्दु c, जिस पर फलन
$f(x) = \log_e\left(\frac{x^2+\alpha}{7x}\right)$, (जहाँ $\alpha \in R$) के लिये अन्तराल [3, 4] में रोल प्रमेय लागु होता है, तो f ″(c) होगा :

(A) $\frac{\sqrt{3}}{7}$ (B) $\frac{1}{12}$

(C) $-\frac{1}{24}$ (D) $-\frac{1}{12}$

12. If the equation
$\frac{b\cos x}{2\cos 2x-1}=\frac{b+\sin x}{(\cos^2 x-3\sin^2 x)\tan x}$,
$b \in R$, has solutions then

(A) $b \in \left(-\infty, \frac{1}{2}\right) - \left\{-1, 0, \frac{1}{3}\right\}$

(B) $b \in (-\infty, 1) - \left\{-1, 0, \frac{1}{3}\right\}$

(C) $b \in R - \left\{-1, 0, \frac{1}{3}\right\}$

(D) $b \in \left(-\infty, \frac{1}{2}\right] - \left\{-1, 0, \frac{1}{3}\right\}$

13. The values of 'k' for which function
$f(x) = \sin x - k(\sin 2x) - \frac{1}{3}(\sin 3x) + 2kx$
increases through out the number line.

(A) $(-\infty, -1)$

(B) $(-\infty, -1]$

(C) $[1, \infty)$

(D) None of these

14. Let altitude AD of acute angled triangle (ΔABC) be produced which meets its circumradius at P such that AP = 4. If circumradius of the triangle is $2\sqrt{2}$ then find the value of $\sec^2(B - C)$.

(A) 3 (B) 2

(C) 4 (D) $2\sqrt{2}$

12. यदि समीकरण
$\frac{b\cos x}{2\cos 2x-1}=\frac{b+\sin x}{(\cos^2 x-3\sin^2 x)\tan x}$,
($b \in R$) का हल हो, तो

(A) $b \in \left(-\infty, \frac{1}{2}\right) - \left\{-1, 0, \frac{1}{3}\right\}$

(B) $b \in (-\infty, 1) - \left\{-1, 0, \frac{1}{3}\right\}$

(C) $b \in R - \left\{-1, 0, \frac{1}{3}\right\}$

(D) $b \in \left(-\infty, \frac{1}{2}\right] - \left\{-1, 0, \frac{1}{3}\right\}$

13. 'k' का मान, जिसके लिये फलन
$f(x) = \sin x - k(\sin 2x) - \frac{1}{3}(\sin 3x) + 2kx$
निरन्तर वर्धमान है, होगा

(A) $(-\infty, -1)$

(B) $(-\infty, -1]$

(C) $[1, \infty)$

(D) इनमें से कोई नहीं

14. माना न्यूनकोण त्रिभुज ABC के शीर्ष लम्ब AD को आगे बढ़ाने पर जो इसकी परिवृत्त को बिन्दु P पर इस प्रकार मिलता है कि AP = 4 है। यदि त्रिभुज की परिवृत्त $2\sqrt{2}$ हो, तो $\sec^2(B - C)$ का मान होगा

(A) 3 (B) 2

(C) 4 (D) $2\sqrt{2}$

15. If $f(x) = x^4 + x^3 + x^2 + x - 1$, then the number of distinct real roots of $f(x) = 0$, is

(A) 0 (B) 1

(C) 2 (D) 4

16. All possible value of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in

(A) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

(B) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

(C) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(D) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$

17. Let $f : R \to R$ be a function such that $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3)$, $x \in R$. Then $f'(2)$ equal :

(A) –10 (B) 15

(C) –6 (D) 10

18. With usual notations in a triangle ABC

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{\lambda(a^2 + b^2 + c^2)R^2}{a^2b^2c^2}$$

Find the value of λ.

(A) 8 (B) 12

(C) 16 (D) 1

15. यदि $f(x) = x^4 + x^3 + x^2 + x - 1$ हो, तो $f(x) = 0$ के विभिन्न वास्तविक मूलों की संख्या होगी

(A) 0 (B) 1

(C) 2 (D) 4

16. $\theta \in [0, 2\pi]$ के सभी सम्भव मान, जिसके लिये $\sin 2\theta + \tan 2\theta > 0$ हो निम्न अन्तराल में स्थित होंगे

(A) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

(B) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

(C) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(D) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$

17. माना $f : R \to R$ एक फलन इस प्रकार है कि $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3)$, $x \in R$ है। तब $f'(2)$ होगा :

(A) –10 (B) 15

(C) –6 (D) 10

18. त्रिभुज ABC में सामान्य संकेतों के साथ

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{\lambda(a^2 + b^2 + c^2)R^2}{a^2b^2c^2}$$

हो, तो λ का मान होगा

(A) 8 (B) 12

(C) 16 (D) 1

19. If

$y = \tan^{-1}\frac{1}{x^2+x+1} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7}$ + ... to n terms, then $\frac{dy}{dx} =$

(A) $\frac{1}{(x+n)^2+1} - \frac{1}{1+x^2}$

(B) $\frac{1}{(x+n-1)^2+1} - \frac{1}{1+x^2}$

(C) $\frac{1}{(x+n)^2+1} - \frac{1}{1+(x+1)^2}$

(D) $\frac{1}{(x+n-1)^2+1} - \frac{1}{1+(x+1)^2}$

20. If p(x) be a polynomial of degree 3 that has a local maximum value 8 at x = 1 and a local minimum value 4 at x = 2; then p(–1) is equal to :

(A) 102

(B) –104

(C) –92

(D) –80

19. यदि

$y = \tan^{-1}\frac{1}{x^2+x+1} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7}$ + ... n पदों तक हो, तो $\frac{dy}{dx} =$

(A) $\frac{1}{(x+n)^2+1} - \frac{1}{1+x^2}$

(B) $\frac{1}{(x+n-1)^2+1} - \frac{1}{1+x^2}$

(C) $\frac{1}{(x+n)^2+1} - \frac{1}{1+(x+1)^2}$

(D) $\frac{1}{(x+n-1)^2+1} - \frac{1}{1+(x+1)^2}$

20. यदि 3 घात का एक बहुपद p(x) इस प्रकार है कि x = 1 पर इसका स्थानीय उच्चिष्ठ मान 8 तथा x = 2 पर स्थानीय निम्निष्ठ मान 4 है, तो p(–1) होगा :

(A) 102

(B) –104

(C) –92

(D) –80

7. Let the function, f : [–7, 0] → R be continuous on [–7, 0] and differentiable on (–7, 0). If f(–7)= –3 and $f'(x) \leq 2$, for all $x \in (-7, 0)$, then for all such functions f(x), the maximum value of $f(-1) + f(0)$ is :

7. माना फलन f : [–7, 0] → R, अन्तराल [–7, 0] में संतत तथा अन्तराल (–7, 0) में अवकलनीय है। यदि f(–7)= –3 तथा $f'(x) \leq 2$, $\forall\, x \in (-7, 0)$ हो, तो सभी ऐसे फलनों f(x) के लिये, $f(-1) + f(0)$ का अधिकतम मान होगा :

8. Let α be the solution of the equation $16^{\sin^2\theta} + 16^{\cos^2\theta} = 10$ in $\left(0, \frac{\pi}{4}\right)$. The angle of elevation of a tower at a distant d meters from its base is angle α. If the tower is 10 meter high, then the value of d is $10\sqrt{\lambda}$, the value of λ is

8. माना $\left(0, \frac{\pi}{4}\right)$ में समीकरण $16^{\sin^2\theta} + 16^{\cos^2\theta} = 10$ का हल α है। इसके आधार से d मीटर की दूरी पर मीनार का उन्नयन का कोण α है। यदि मीनार की ऊँचाई 10 मीटर है, तब d का मान $10\sqrt{\lambda}$ हो, तो λ का मान हेगा

9. $f(x) = \prod_{n=1}^{100} (x - n)^{n(101-n)}$, find $\frac{f'(101)}{f(101)} = ?$

9. $f(x) = \prod_{n=1}^{100} (x - n)^{n(101-n)}$ हो, तो $\frac{f'(101)}{f(101)} = ?$

10. Number of integral solution(s) of the inequality $4\sin^2 x - 8\sin x + 3 \leq 0$, in $x \in [0, 2\pi]$ is

10. $x \in [0, 2\pi]$ में असमिका $4\sin^2 x - 8\sin x + 3 \leq 0$ के पूर्णांक हलों की संख्या होगी

Space for Rough Work / रफ कार्य के लिए जगह

8. **Ans (B)**

$V = 2f_0 (l_2 - 1_1)$

$(dV) = 2f_0 (dl_2 - dl_1)$

$(dV)_{max}$ = max of $[2f_0(\pm \Delta l_2 \pm \Delta l_1]$

$= 2f_0 (\Delta l_2 + \Delta l_1)$

$l_1 = 25.0$ cm $\Rightarrow \Delta l_1 = 0.1$ cm (place value of last number)

$l_2 = 75.0$ cm $\Rightarrow \Delta l_2 = 0.1$ cm (place value of last number)

So the maximum permissible error in the speed of sound $(dV)_{max}$

$= 2(325Hz)\ (0.1\ cm + 0.1\ cm) = 1.3\ m\ s^{-1}$

9. **Ans (B)**

$34.123 + 2.10 = 36.223 = 36.22$

$98.7654 - 4.983 = 93.7824 = 93.782$

$\frac{36.22}{93.782} = 0.386214... = 0.3862$

(lowest significant value)

10. **Ans (B)**

$\frac{R_x}{R} = \frac{x}{L-x}$

$R_x = R\left(\frac{x}{L-x}\right)$

For $R = 20\Omega$, $x = 20$ cm

$R_x = 20\left[\frac{20}{100-20}\right] = 5\Omega$

from (i) $\ln R_x = \ln r + \ln x - \ln (L - x)$

$\frac{dR_x}{R_x} = \frac{dx}{x} + \frac{dx}{L-x}$

$\frac{\Delta R_x}{R_x} = \frac{L\Delta x}{x(L-x)}$ [Δx = maximum error in x]

for $R_x = 5\Omega$; $|\Delta x| = 0.1$ cm;

$L = 100$ cm $x = 20$ cm

$\therefore |\Delta R_x| = \frac{5 \times 100 \times 0.1}{20 \times 80} = 0.03\Omega$

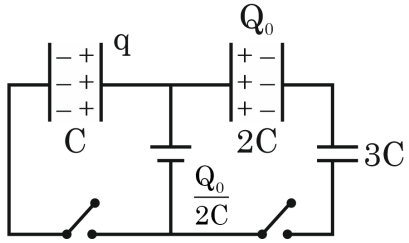
$\therefore R_x = (5.00 \pm 0.03)\Omega$

11. **Ans (C)**

$= \frac{2\varepsilon_0 A}{5d} + \frac{2\varepsilon_0 A}{5\left(\frac{3d}{4}\right)} + \frac{2\varepsilon_0 A}{5\,(d)} = \frac{4\varepsilon_0 A}{3d}$

12. **Ans (A)**

After closing the switches.



$q = \frac{Q_0}{2c} \times C$

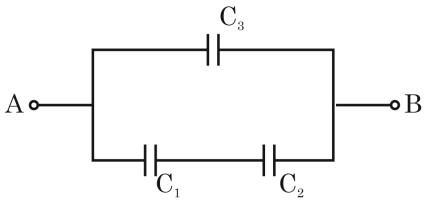
$\frac{Q_0}{2}$

$\therefore$ charge flown through

$S_2 = \frac{Q_0}{2} - \frac{Q_0}{3} = \frac{Q_0}{6}$

charge flown through $S_1 = 0$

13. **Ans (C)**



$C = \frac{\varepsilon_0 A}{d}$ assume

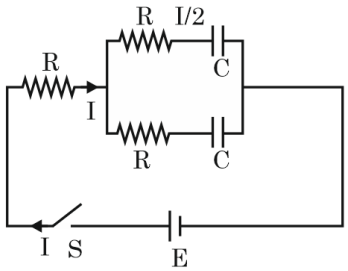
$C_1 = \frac{2\varepsilon_0 A}{d} = 2C$

$C_2 = \frac{3\varepsilon_0 A}{d} = 3C$

$C_3 = \frac{\varepsilon_0 A}{2d} = \frac{C}{2}$

$C_{AB} = \frac{C_1 C_2}{C_1 + C_2} + C_3 = \frac{17\varepsilon_0 A}{10d}$

14. **Ans (C)**



Applying KVL, we get

$\frac{I}{2} = \frac{dq}{dt}$ or $I = 2\frac{dq}{dt}$

$E - IR - \frac{I}{2}R - \frac{q}{C} = 0$

or $E - \frac{3}{2}IR - \frac{q}{C} = 0$

or $E - \frac{3}{2} \times 2\frac{dq}{dt}R - \frac{q}{C} = 0$

$q_{(t)} = EC\,(1 - e^{-t/3RC})$

Score-I - 28 Jan 2022

ANSWER KEY

PAPER-1

PHYSICS

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	A	B	C	A	D	C,D	A,C	B,C,D	A,C,D

Q11	Q12	Q1	Q2	Q3	Q4	Q5	Q6
A,B,C	B	9.64	11.00	9.00	22.50	1.75	19.00

CHEMISTRY

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	D	B	B	C	B	A,C,D	C	A,D	A

Q11	Q12	Q2	Q3	Q4	Q5	Q6
A,C,D	A,B,C	250.00	6.00	5.00	3.00	420.00

MATHEMATICS

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	A	C	C	B	D	A,B,C	A,B	A,B	A,C,D

Q11	Q12	Q1	Q2	Q3	Q4	Q5	Q6
B,C	A,B,C,D	1.00	4.00	1.00	1.50	8.00	43.00

PAPER-2

PHYSICS

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	D	A	C	D	A	D	B	B	B

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
C	A	C	C	A	C	A	A	C	A

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
14.00	251.00	1.00	2.00	37.50	2.60	2.50	54.00	8.00	594.00

CHEMISTRY

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	D	C	C	B	C	B	B	A	C

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
D	C	B	B	B	D	D	B	A	A

Q1	Q2	Q3	Q4	Q5	Q6	Q8	Q9	Q10
1.00	4.00	1.00	4.00	3.47	5.00	2.00	51.00	9.00

MATHEMATICS

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	D	A	A	B	D	A	D	C	B

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
B	A	C	B	C	B	C	C	A	B

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
63.50	6.00	A	C	D	A	D	3.00	5050.00	2.00

Q15	Q16	Q17	Q18	Q19	Q20
A	C	A	A	C	A